

Coupling plant physiology and pest demography to understand plant-nematode interactions

Joseph Penlap, Suzanne Touzeau, Valentina Baldazzi, Frédéric Grogard

INRAE / UMR 1355 ISA / M2P2 & Inria / MACBES

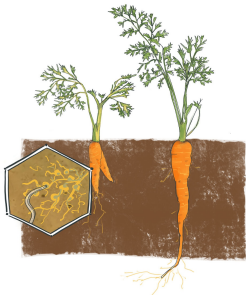
Spring conference, March 10-14, 2025, Sophia Antipolis, France

Population dynamics: "From Data to Models"



Root-Knot Nematodes (RKN), *Meloidogyne* spp.

- small soil worms,
- obligate root endoparasites,
- ubiquitous polyphagous pest
- 14% of global crop losses worldwide [1]
[1] Djian-Caporalino, *EPPO Bulletin*, 2012

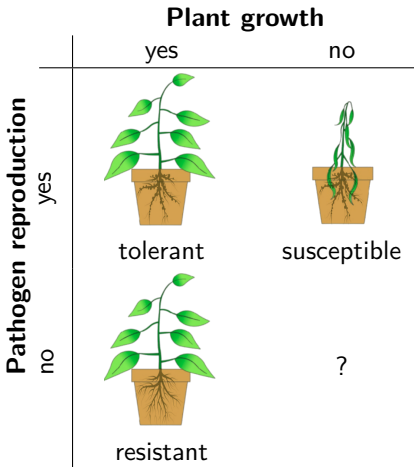


Main impacts

- root deformation (galls)
- stunted growth and wilting

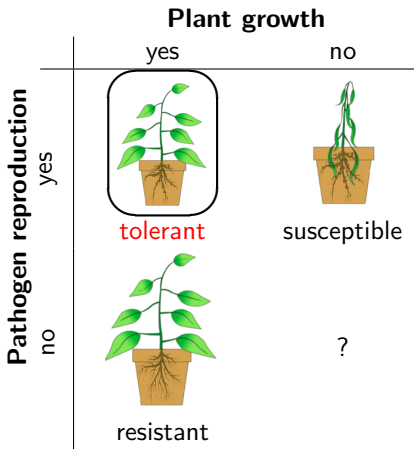
Plant variability

Strong variability in plant response to RKN parasitism among crop species



Plant variability

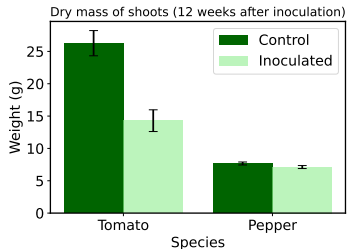
Strong variability in plant response to RKN parasitism among crop species



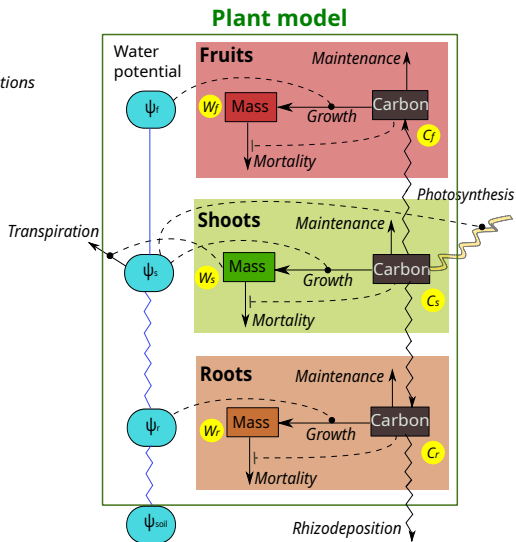
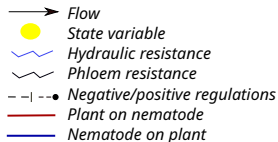
Which
mechanisms
underlie plant
tolerance?

Experimental data (ArchiNem project)

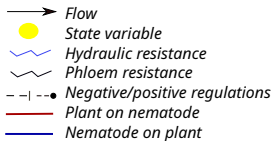
- 2 plant species: tomato, pepper
- 2 plant categories:
 - control
 - inoculated by nematodes
- 3 points in time (T0, T6 = 6 wai, T12 = 12 wai)
- 6 replicates / category / point
- weekly number of leaves
- destructive measures:
 - **plant**: fresh and dry masses for shoot, root and fruit
 - **nematodes**:
 - number of galls (T6)
 - egg masses (T6, T12)
 - egg per egg masses (T6, T12)
 - soil-extracted larvae (T12)



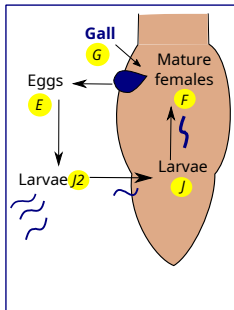
Integrated plant-pest model



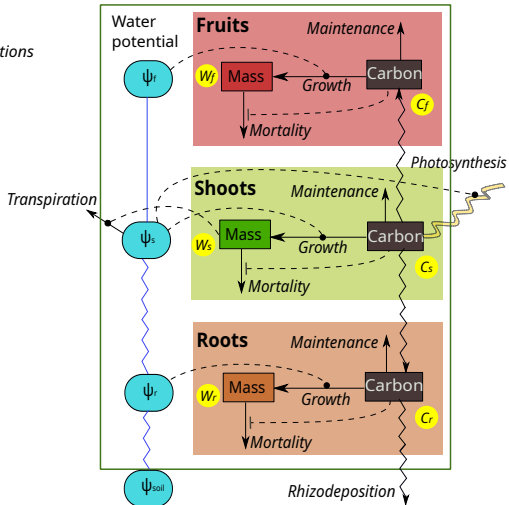
Integrated plant-pest model



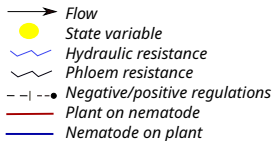
Nematode model



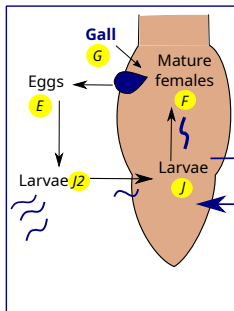
Plant model



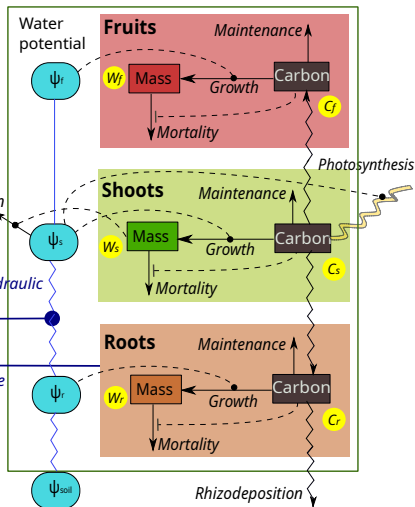
Integrated plant-pest model



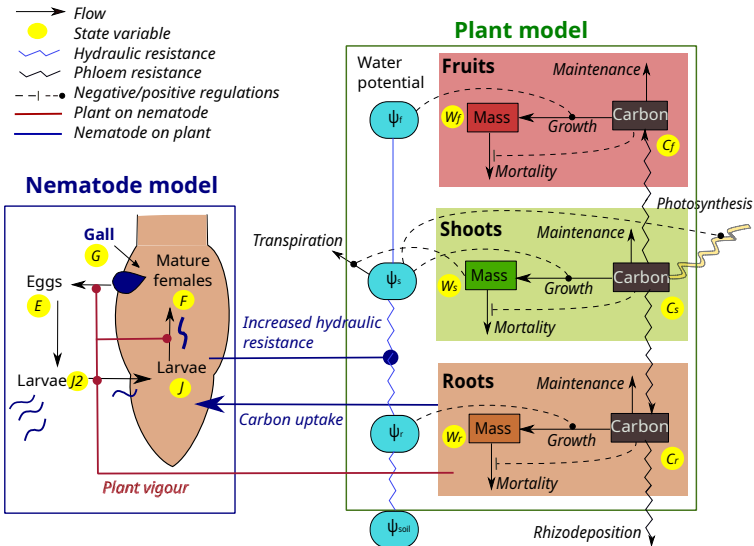
Nematode model



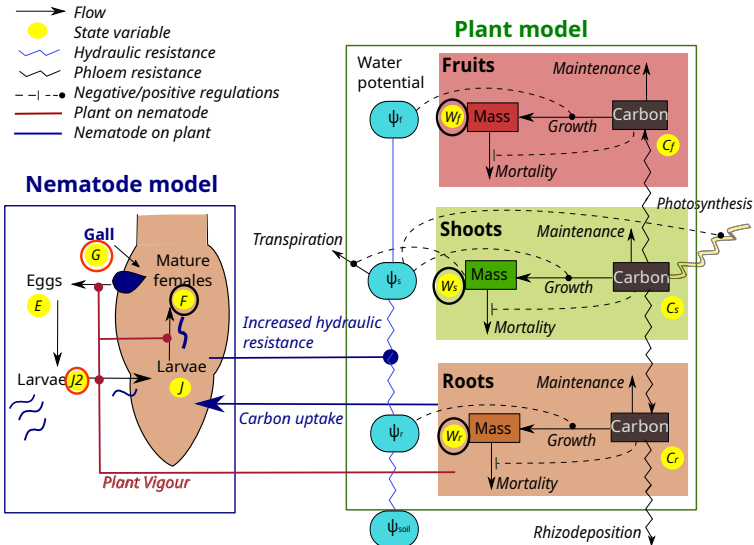
Plant model



Integrated plant-pest model



Integrated plant-pest model



+ weekly number of leaves

Mathematical equations: plant model

$$\text{Shoot} \left\{ \begin{array}{l} \frac{dW_s}{dt} = \\ \frac{dC_s}{dt} = \end{array} \right.$$

$$\text{Root} \left\{ \begin{array}{l} \frac{dW_r}{dt} = \\ \frac{dC_r}{dt} = \end{array} \right.$$

$$\text{Fruit} \left\{ \begin{array}{l} \frac{dW_f}{dt} = \\ \frac{dC_f}{dt} = \end{array} \right.$$

Mathematical equations: plant model

$$\text{Shoot} \left\{ \begin{array}{l} \frac{dW_s}{dt} = \\ \frac{dC_s}{dt} = \underbrace{p \left(\frac{1}{K_u + W_s} \right) f_p(\psi_s)}_{\text{Uptake}} \end{array} \right.$$

$$- \underbrace{\frac{1}{W_s} \frac{dW_s}{dt} C_s}_{\text{Dilution}}$$

$$\text{Root} \left\{ \begin{array}{l} \frac{dW_r}{dt} = \\ \frac{dC_r}{dt} = \end{array} \right.$$

$$- \underbrace{\frac{1}{W_r} \frac{dW_r}{dt} C_r}_{\text{Dilution}}$$

$$\text{Fruit} \left\{ \begin{array}{l} \frac{dW_f}{dt} = \\ \frac{dC_f}{dt} = \end{array} \right.$$

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Mathematical equations: plant model

$$\begin{aligned}
 &\text{Shoot} \left\{ \begin{aligned} \frac{dW_s}{dt} &= \\ \frac{dC_s}{dt} &= \underbrace{p \left(\frac{1}{K_u + W_s} \right) f_p(\psi_s)}_{\text{Uptake}} - \underbrace{\frac{1}{W_s} (T_r + T_f + T_a)}_{\text{Transport}} - \underbrace{\frac{1}{W_s} \frac{dW_s}{dt} C_s}_{\text{Dilution}} \end{aligned} \right. \\
 &\text{Root} \left\{ \begin{aligned} \frac{dW_r}{dt} &= \\ \frac{dC_r}{dt} &= \frac{1}{W_r} \underbrace{T_r}_{\text{Transport}} - \underbrace{\frac{1}{W_r} \frac{dW_r}{dt} C_r}_{\text{Dilution}} \end{aligned} \right. \\
 &\text{Fruit} \left\{ \begin{aligned} \frac{dW_f}{dt} &= \\ \frac{dC_f}{dt} &= \frac{1}{W_f} \underbrace{(T_f + T_a)}_{\text{Transport}} - \underbrace{\frac{1}{W_f} \frac{dW_f}{dt} C_f}_{\text{Dilution}} \end{aligned} \right.
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Mathematical equations: plant model

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 \text{Shoot} \quad & \left\{ \begin{aligned} \frac{dW_s}{dt} &= \underbrace{k_s f_g(\psi_s) \frac{C_s}{K_s + C_s} W_s}_{\text{Growth}} \\ \frac{dC_s}{dt} &= \underbrace{p \left(\frac{1}{K_u + W_s} \right) f_p(\psi_s)}_{\text{Uptake}} - \underbrace{\frac{1}{W_s} (T_r + T_f + T_a)}_{\text{Transport}} - \underbrace{(b) k_s f_g(\psi_s) \frac{C_s}{K_s + C_s}}_{\text{Growth}} - \underbrace{\frac{1}{W_s} \frac{dW_s}{dt} C_s}_{\text{Dilution}} \end{aligned} \right. \\
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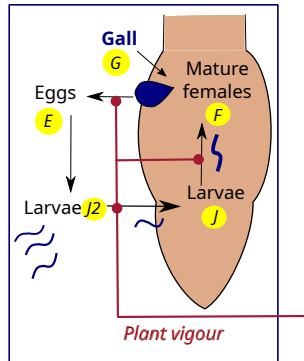
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 \end{aligned}$$

Plant status on RKN development

$$\begin{cases}
 \frac{dE}{dt} = \underbrace{\rho(C_r) F}_{\text{Egg laying}} - \underbrace{\lambda E}_{\text{Hatching}} - \underbrace{\mu_E E}_{\text{Mortality}}, \\
 \frac{dJ_2}{dt} = \underbrace{\lambda E}_{\text{Hatching}} - \underbrace{\beta J_2 W_r}_{\text{Root entering}} - \underbrace{\mu_{J_2} J_2}_{\text{Mortality}}, \\
 \frac{dJ}{dt} = \underbrace{\omega(C_r) \beta J_2 W_r}_{\text{Entry + establishment}} - \underbrace{\eta J}_{\text{Maturation}} - \underbrace{(\mu_J + \nu(C_r)) J}_{\text{Mortality}}, \\
 \frac{dF}{dt} = \underbrace{\eta J}_{\text{Maturation}} - \underbrace{\mu_F F}_{\text{Mortality}}.
 \end{cases}$$

Nematode model



RKN effects on plant growth

$$\text{Root} \left\{ \begin{aligned}
 \frac{dW_r}{dt} &= \underbrace{k_r f_g(\psi_r) \frac{C_r}{K_r + C_r} W_r}_{\text{Growth}} - \underbrace{d_r \left(1 + \frac{K_m^n}{K_m^n + C_r^n} \right) W_r}_{\text{Mortality}} \\
 \frac{dC_r}{dt} &= \frac{1}{(W_r + G)} \underbrace{T_r}_{\text{Transport}} - \underbrace{(b + r_{g,r}) \frac{f_g(\psi_r)}{W_r + G} \left(k_r \frac{C_r}{K_r + C_r} W_r + k_g \frac{C_r}{(K_r/v) + C_r} G \right)}_{\text{Growth + Respiration}} - \underbrace{r_{m,r} \left(\frac{C_r^n}{K_m^n + C_r^n} \right)}_{\text{Maintenance}} \\
 &\quad - \underbrace{z C_r}_{\text{Rhizodeposition}} - \underbrace{\frac{C_r}{(W_r + G)} \left(\frac{dW_r}{dt} + \frac{dG}{dt} \right)}_{\text{Dilution}}
 \end{aligned} \right.$$

RKN effects on plant growth

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 \frac{dC_r}{dt} &= \frac{1}{(W_r + G)} \underbrace{T_r}_{\text{Transport}} - \underbrace{(b + r_{g,r}) \frac{f_g(\psi_r)}{W_r + G} \left(k_r \frac{C_r}{K_r + C_r} W_r + k_g \frac{C_r}{(K_r/v) + C_r} G \right)}_{\text{Growth + Respiration}} - \underbrace{r_{m,r} \left(\frac{C_r^n}{K_m^n + C_r^n} \right)}_{\text{Maintenance}} \\
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RKN effects on plant growth

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 \frac{dG}{dt} &= \underbrace{\omega(C_r) \epsilon \beta J_2 W_r}_{\text{Gall formation}} + \underbrace{k_g f_g(\psi_r) \frac{C_r}{(K_r/v) + C_r} G}_{\text{Growth}} - \underbrace{d_r G}_{\text{Natural mortality}} \\
 \frac{dC_r}{dt} &= \frac{1}{(W_r + G)} \underbrace{T_r}_{\text{Transport}} - \underbrace{(b + r_{g,r}) \frac{f_g(\psi_r)}{W_r + G} \left(k_r \frac{C_r}{K_r + C_r} W_r + k_g \frac{C_r}{(K_r/v) + C_r} G \right)}_{\text{Growth + Respiration}} - \underbrace{r_{m,r} \left(\frac{C_r^n}{K_m^n + C_r^n} \right)}_{\text{Maintenance}} \\
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RKN effects on plant growth

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 \frac{dC_r}{dt} &= \frac{1}{(W_r + G)} \underbrace{T_r}_{\text{Transport}} - \underbrace{(b + r_{g,r}) \frac{f_g(\psi_r)}{W_r + G} \left(k_r \frac{C_r}{K_r + C_r} W_r + k_g \frac{C_r}{(K_r/v) + C_r} G \right)}_{\text{Growth + Respiration}} - \underbrace{r_{m,r} \left(\frac{C_r^n}{K_m^n + C_r^n} \right)}_{\text{Maintenance}} \\
 &\quad - \underbrace{z C_r}_{\text{Rhizodeposition}} - \underbrace{\frac{C_r}{(W_r + G)} \left(\frac{dW_r}{dt} + \frac{dG}{dt} \right)}_{\text{Dilution}} - \underbrace{\frac{1}{(W_r + G)} \gamma \left(\frac{C_r}{K_\gamma + C_r} \right) (F + J)}_{\text{Nematode feeding}}.
 \end{aligned} \right.$$

Calibration procedure

Two-phase strategy

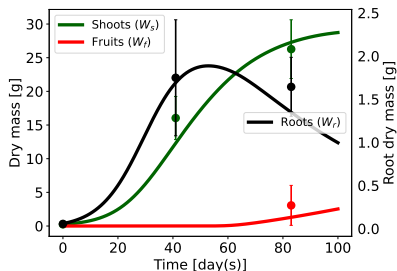
- calibration of plant model on control data
- calibration of plant-RKN model on inoculated plants & RKN data

Steps

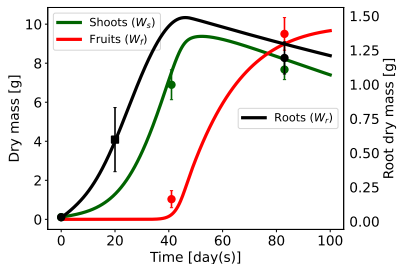
- select parameters to be estimated based on **sensitivity analysis**
 - extensive literature review
 - 8 parameters estimated out of a total of 27
- least square **criterion** between model simulations and data
- optimisation
 - Global search: ARS (Adaptive Random Search)
 - Local search: Nelder-Mead method

Plant model calibration

Healthy tomato dynamics



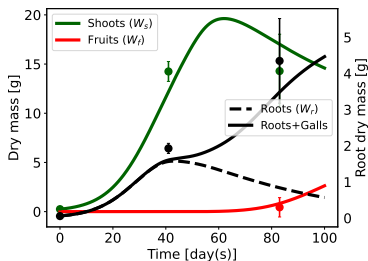
Healthy pepper dynamics



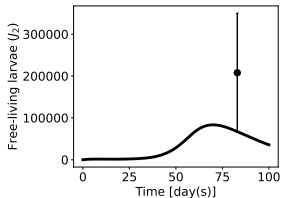
Good agreement with data

Plant-RKN model calibration: tomato

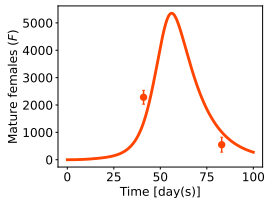
Infested tomato dynamics



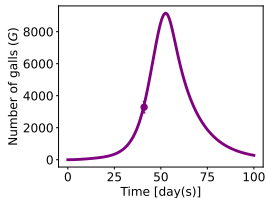
Juvenile larvae



Females

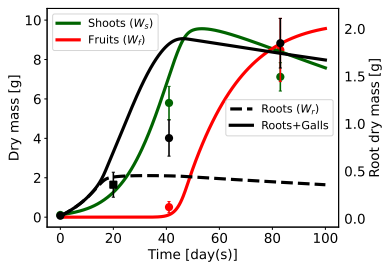


Galls

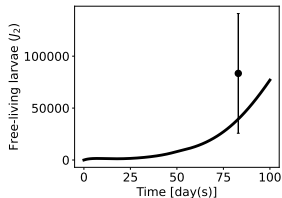


Plant-RKN model calibration: pepper

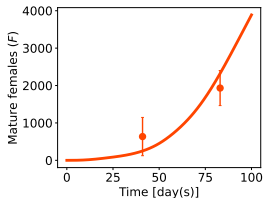
Infested pepper dynamics



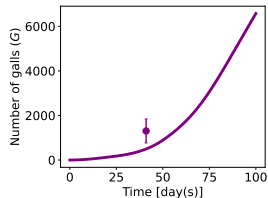
Juvenile larvae



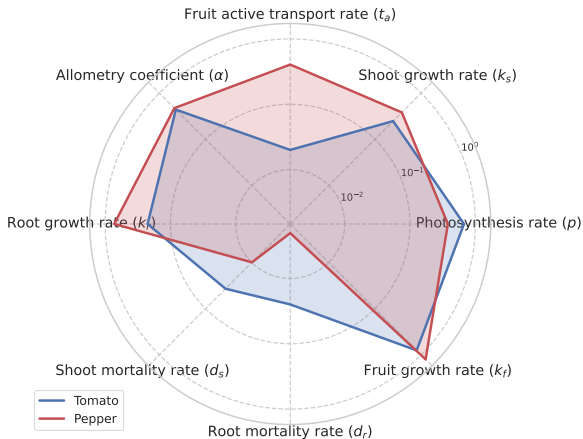
Females



Galls



Tomato vs Pepper: Plant parameters



Pepper: reduced mortality γ_r , higher growth k_r , faster C allocation towards fruits t_a

Virtual tolerant tomato

Are these traits really important for plant tolerance?

Virtual tolerant tomato

Are these traits really important for plant tolerance?

Pepper light tolerance?

- reduced root mortality rate (d_r)
- higher root growth rate (k_r)
- faster carbon allocation towards fruits (t_a)



All other parameters are kept fixed to estimated values

Virtual tolerant tomato

Are these traits really important for plant tolerance?

Pepper light tolerance?

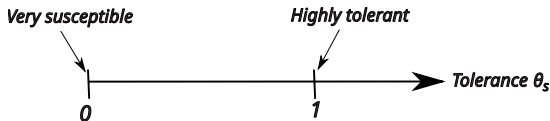
- reduced root mortality rate (d_r)
- higher root growth rate (k_r)
- faster carbon allocation towards fruits (t_a)



All other parameters are kept fixed to estimated values

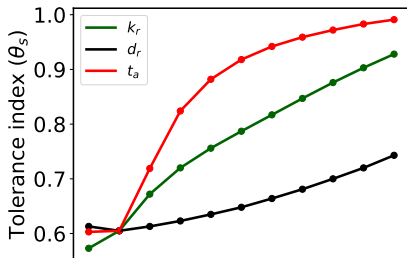
Tolerance index (over the cropping season $T = 200\text{days}$)

$$\theta_s = \frac{\int_0^T W_s^{\text{inoc}}(t) dt}{\int_0^T W_s^{\text{healthy}}(t) dt} = \frac{\text{Cumulative dry mass of } \mathbf{infected} \text{ shoots}}{\text{Cumulative dry mass of } \mathbf{healthy} \text{ shoots}}$$

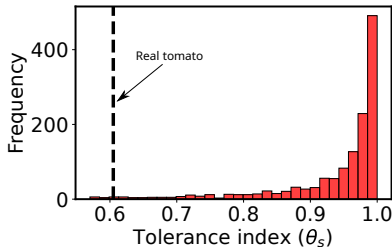


Building virtual tomato: pepper inspired traits

Variation of one parameter at a time



Full parameter combinations



Conclusion : more tolerant tomatoes

Virtual tomato: other tolerance traits?

Estimated parameters : Carbon allocation towards fruits (t_a),

Root growth (k_r), Root mortality (d_r), Shoot growth (k_s), Shoot mortality (d_s), Photosynthesis (p), Fruit growth (k_f), Allometry coefficient (α)

Fixed parameters : Phloem resistance ($r_{ph,r}$), Hydraulic resistance ($r_{xy,r}$)

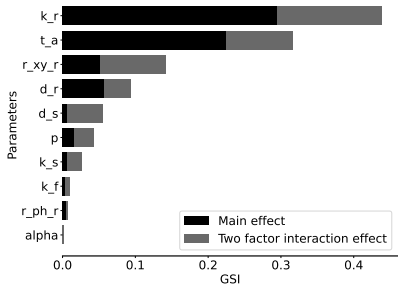
Virtual tomato: other tolerance traits?

Estimated parameters : Carbon allocation towards fruits (t_a),

Root growth (k_r), Root mortality (d_r), Shoot growth (k_s), Shoot mortality (d_s), Photosynthesis (p), Fruit growth (k_f), Allometry coefficient (α)

Fixed parameters : Phloem resistance ($r_{ph,r}$), Hydraulic resistance ($r_{xy,r}$)

Sensitivity analysis on θ_s



Conclusion : First 4 parameters strongly impact the tolerance index.

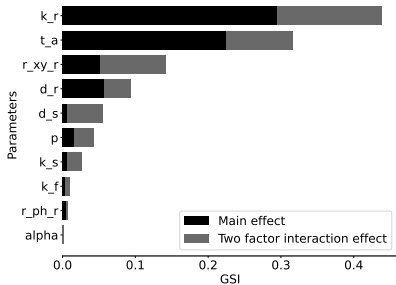
Virtual tomato: other tolerance traits?

Estimated parameters : Carbon allocation towards fruits (t_a),

Root growth (k_r), Root mortality (d_r), Shoot growth (k_s), Shoot mortality (d_s), Photosynthesis (p), Fruit growth (k_f), Allometry coefficient (α)

Fixed parameters : Phloem resistance ($r_{ph,r}$), Hydraulic resistance ($r_{xy,r}$)

Sensitivity analysis on θ_s



Plant tolerance mechanisms

- high root growth (k_r)
- low root mortality (d_r)
- faster carbon allocation towards fruits (t_a)
- low **hydraulic resistance** ($r_{xy,r}$)

Conclusion : First 4 parameters strongly impact the tolerance index.

Conclusion & Perspectives

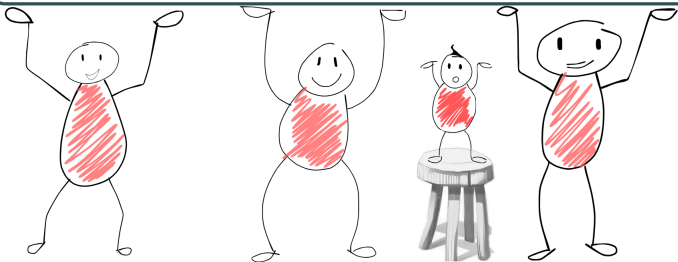
Summary

- Mechanistic model of plant-RKN interactions
 - coupling plant physiology & pest demography
 - better prediction of plant growth and yield
- Model calibration on experimental data
 - good agreement with data
- Comparison of species traits (tomato & pepper)
 - screening of interesting plant tolerance traits
 - in silico validation of tolerance traits

Perspectives

- Robustness of tolerance traits
 - infection scenarios (low, medium, high RKN virulence)
 - time horizon (multi-seasonal)

THANK YOU FOR YOUR ATTENTION!



Collaborators:

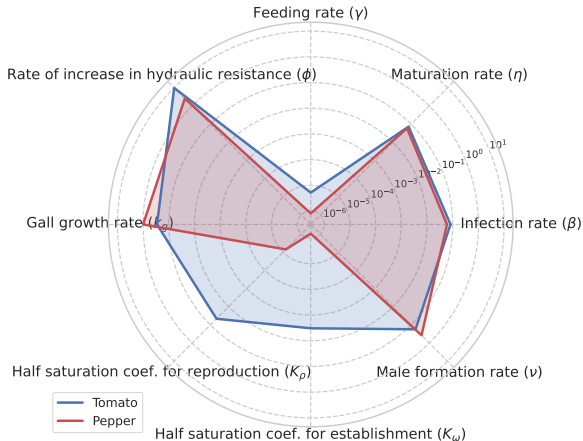
INRAE colleagues

- C. Caporalino
- L. Pagès
- C. Doussan

Master students

- T. Brenière
- N. Juazion-Graverolle
- C. Bourgade
- E. Ceci
- A. Canaud

Tomato vs Pepper: RKN infection parameters



Tomato: Higher effect of plant status on RKN reproduction K_ρ and establishment K_ω rates

Tolerance Index

Yield

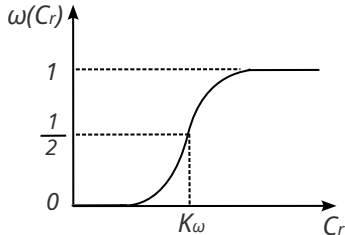
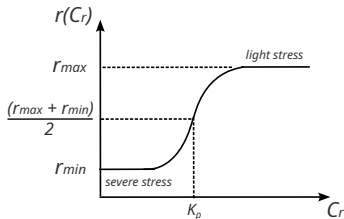
$$\theta_s = \frac{\int_0^T W_s^{inoc}(t) dt}{\int_0^T W_s^{healthy}(t) dt},$$

Plant Model: Mathematical equations

$$\begin{aligned}
 \text{Shoot} \left\{ \begin{aligned} \frac{dW_s}{dt} &= \underbrace{k_s f_g(\psi_s) \frac{C_s}{K_s + C_s} W_s}_{\text{Growth}} - \underbrace{d_s \left(1 + \frac{K_m^n}{K_m^n + C_s^n} \right) W_s}_{\text{Mortality}}, \\ \frac{dC_s}{dt} &= \underbrace{p \left(\frac{1}{K_u + W_s} \right) f_p(\psi_s)}_{\text{Uptake}} - \underbrace{\frac{1}{W_s} (T_r + T_f + T_a)}_{\text{Transport}} - \underbrace{(b + r_{g,s}) k_s f_g(\psi_s) \frac{C_s}{K_s + C_s}}_{\text{Growth+Respiration}} - \underbrace{r_{m,s} \left(\frac{C_s^n}{K_m^n + C_s^n} \right)}_{\text{Maintenance}} - \underbrace{\frac{1}{W_s} \frac{dW_s}{dt} C_s}_{\text{Dilution}}, \end{aligned} \right. \\
 \text{Root} \left\{ \begin{aligned} \frac{dW_r}{dt} &= \underbrace{k_r f_g(\psi_r) \frac{C_r}{K_r + C_r} W_r}_{\text{Growth}} - \underbrace{d_r \left(1 + \frac{K_m^n}{K_m^n + C_r^n} \right) W_r}_{\text{Mortality}}, \\ \frac{dC_r}{dt} &= \underbrace{\frac{1}{W_r} T_r}_{\text{Transport}} - \underbrace{(b + r_{g,r}) k_r f_g(\psi_r) \frac{C_r}{K_r + C_r}}_{\text{Growth+Respiration}} - \underbrace{r_{m,r} \left(\frac{C_r^n}{K_m^n + C_r^n} \right)}_{\text{Maintenance}} - \underbrace{z C_r}_{\text{Rhizodeposition}} - \underbrace{\frac{1}{W_r} \frac{dW_r}{dt} C_r}_{\text{Dilution}}, \end{aligned} \right. \\
 \text{Fruit} \left\{ \begin{aligned} \frac{dW_f}{dt} &= \underbrace{k_f f_g(\psi_f) \frac{C_f}{K_f + C_f} W_f}_{\text{Growth}} - \underbrace{d_f \left(1 + \frac{K_m^n}{K_m^n + C_f^n} \right) W_f}_{\text{Mortality}}, \\ \frac{dC_f}{dt} &= \underbrace{\frac{1}{W_f} (T_f + T_a)}_{\text{Transport}} - \underbrace{(b + r_{g,f}) k_f f_g(\psi_f) \frac{C_f}{K_f + C_f}}_{\text{Growth+Respiration}} - \underbrace{r_{m,f} \left(\frac{C_f^n}{K_m^n + C_f^n} \right)}_{\text{Maintenance}} - \underbrace{\frac{1}{W_f} \frac{dW_f}{dt} C_f}_{\text{Dilution}}. \end{aligned} \right.
 \end{aligned}$$

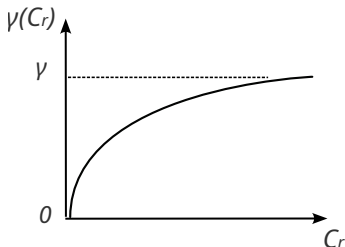
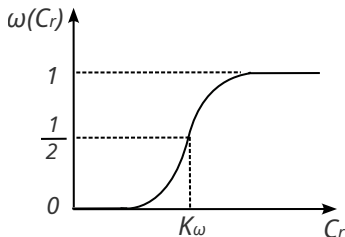
Plant status on RKN development

$$\begin{aligned}
 \text{Free} \quad & \left\{ \begin{aligned} \frac{dE}{dt} &= \underbrace{r(C_r) F}_{\text{Egg laying}} - \underbrace{h E}_{\text{Egg hatching}} - \underbrace{\mu_e E}_{\text{Mortality}} \\ \frac{dJ_2}{dt} &= \underbrace{h E}_{\text{Egg hatching}} - \underbrace{\beta J_2 W_r}_{\text{Larvae infection}} - \underbrace{\mu_{J_2} J_2}_{\text{Mortality}} \end{aligned} \right. \\
 \text{RKN} \quad & \left\{ \begin{aligned} \frac{dJ}{dt} &= \underbrace{\Omega(C_r) \beta J_2 W_r}_{\text{RKN entry}} - \underbrace{\eta J}_{\text{Maturation}} - \underbrace{\mu_j J}_{\text{Mortality}} \\ \frac{dF}{dt} &= \underbrace{\eta J}_{\text{Maturation}} - \underbrace{\mu_F F}_{\text{Mortality}} \end{aligned} \right.
 \end{aligned}$$



RKN effects on plant growth

$$\begin{aligned}
 \left. \begin{aligned}
 \frac{dW_r}{dt} &= \underbrace{k_r f(\psi_r) \frac{C_r}{K_r + C_r} W_r}_{\text{Growth}} - \underbrace{\gamma_r \left(1 + \frac{K_m^n}{K_m^n + C_r^n}\right) W_r}_{\text{Mortality}} - \underbrace{\Omega(C_r) \epsilon \beta J_2 W_r}_{\text{Successfully infected roots}} \\
 \frac{dG}{dt} &= \underbrace{\Omega(C_r) \epsilon \beta J_2 W_r}_{\text{Gall formation}} + \underbrace{k_g f(\psi_r) \frac{C_r}{K_r + C_r} G}_{\text{Growth}} - \underbrace{\Gamma_r G}_{\text{Natural mortality}} \\
 \frac{dC_r}{dt} &= \underbrace{\frac{1}{(W_r + G)} T_r}_{\text{Transport}} - \underbrace{(f_c + r_{g,r}) \left(\frac{k_r W_r + k_g G}{W_r + G}\right) f(\psi_r) \frac{C_r}{K_r + C_r}}_{\text{Growth}} - \underbrace{r_{m,r} \left(\frac{C_r^n}{K_m^n + C_r^n}\right)}_{\text{Maintenance respiration}} - \underbrace{c_{rh} C_r}_{\text{Rhizodeposition}} \\
 &\quad - \underbrace{\frac{C_r}{(W_r + G)} \left(\frac{dW_r}{dt} + \frac{dG}{dt}\right)}_{\text{dilution}} - \underbrace{\frac{1}{(W_r + G)} \gamma(C_r) (F + J)}_{\text{Nematode feeding}}
 \end{aligned} \right\} \text{Root}
 \end{aligned}$$



Global sensitivity analysis (GSA)

Goal: Identifying the most influential parameters

inputs : all parameters, **outputs** : model variables along time (vector)

Steps

1. fractional factorial design to explore parameter space
2. PCA to reduce output and capture its variability
3. ANOVA to compute sensitivity indices (SI)

$$SI_{..} = \frac{SS_{..}}{TSS}, \quad GSI = \sum_{k=1}^{components} SI_k \times inertia_k$$

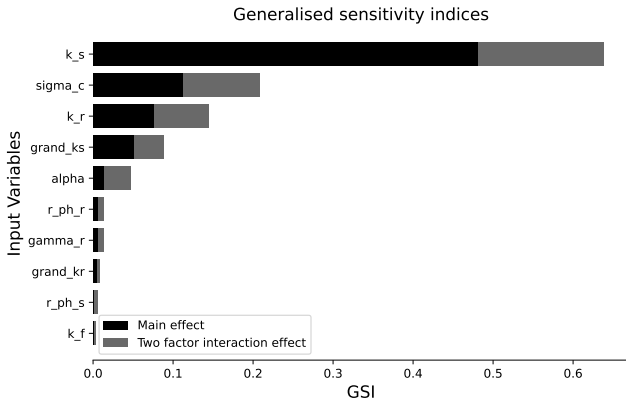
SS = sum of squares, TSS = total sum of squares

Global sensitivity analysis on plant model

Features

- **goal:** identify the most influential parameters
- **inputs:** 26 parameters, **outputs:** W_s , W_r , W_f , C_s , C_r , C_f

Plant traits that most affect **root biomass** (W_r) dynamics

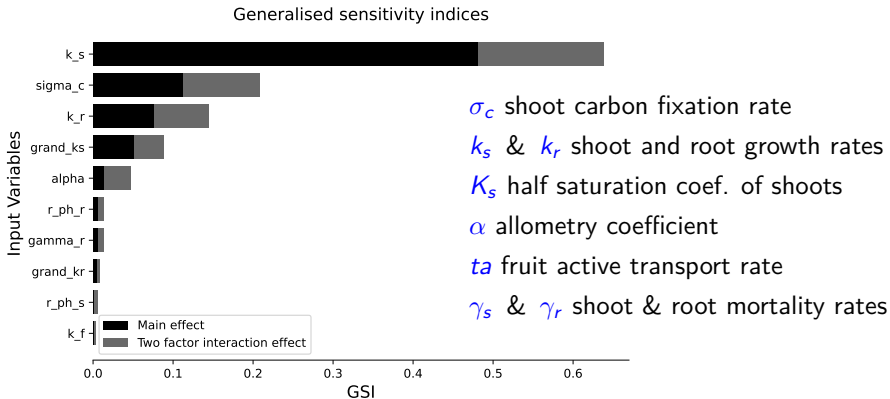


Global sensitivity analysis on plant model

Features

- **goal:** identify the most influential parameters
- **inputs:** 26 parameters, **outputs:** W_s , W_r , W_f , C_s , C_r , C_f

Plant traits that most affect **root biomass** (W_r) dynamics



Data processing

Objective: increase the number of data points by exploiting non-destructive measurements (leaf number)

Allometric relation: shoot weight vs leaf number

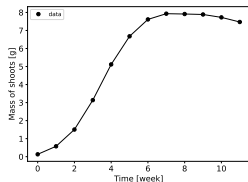
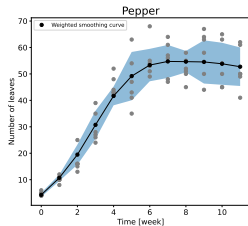
- Spline smoothing of leaf data
 - average dynamics
- Calibration of a general allometric relation (3 time-points x 3 species)

$$M = a L^{b_i}, \quad i = \text{species } 1, 2, 3$$

M = shoot dry mass, L = number of leaves



- Prediction of shoot dry mass from leaf data at 12 time-points



Criterion for estimating interaction parameters

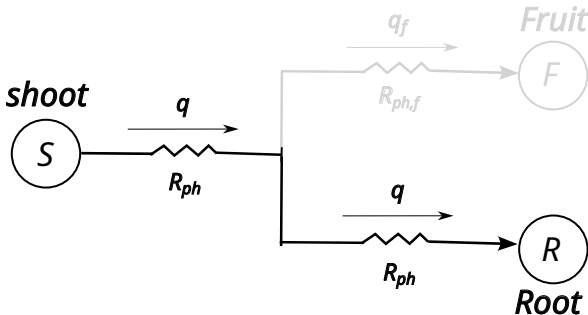
Cost function

$$err = \underbrace{\alpha_c \left[\frac{1}{2} \sum_{i=1}^2 \left(\frac{(W_s^{sol})_i - (W_s^{obs})_i}{(W_s^{obs})_i} \right)^2 + \frac{1}{2} \sum_{i=1}^2 \left(\frac{(W_r^{sol})_i - (W_r^{obs})_i}{(W_r^{obs})_i} \right)^2 + \left(\frac{W_f^{sol} - W_f^{obs}}{W_f^{obs}} \right)^2 \right]}_{\text{Inoculated plant}} + \underbrace{\beta_c \left[\left(\frac{G^{sol} - G^{obs}}{G^{obs}} \right)^2 + \frac{1}{2} \sum_{i=1}^2 \left(\frac{F_i^{sol} - F_i^{obs}}{F_i^{obs}} \right)^2 + \frac{1}{2} \sum_{i=1}^2 \left(\frac{r_i^{sol} - r_i^{obs}}{r_i^{obs}} \right)^2 \right]}_{\text{Nematodes}}$$

- α_c and $\beta_c = (1 - \alpha_c)$ are the weighted coefficients
- G^{sol} , F^{sol} , r^{sol} are nematode model predictions of galls, females and egg-laying
- G^{obs} , F^{obs} , r^{obs} are nematode data

Carbon transport (T)

The carbon flow³ T in phloem vessels $T = q C_s$, with q the volume flow rate



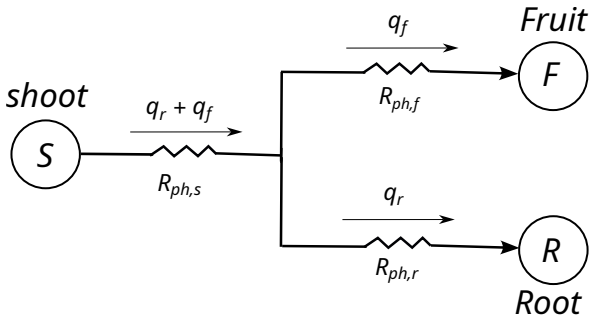
Then,

$$\begin{cases} (q_r + q_f)R_{ph,s} + q_r R_{ph,r} = (C_s - C_r) \\ (q_r + q_f)R_{ph,s} + q_f R_{ph,f} = (C_s - C_f) \end{cases}$$

$$\begin{cases} T = \frac{(C_s - C_r)}{R_{ph}} C_s \\ R_{ph,..} = \frac{r_{ph,..}}{W_{..}^{\alpha..}} \end{cases}$$

³Minchin et al., *Journal of Experimental Botany*, 1993

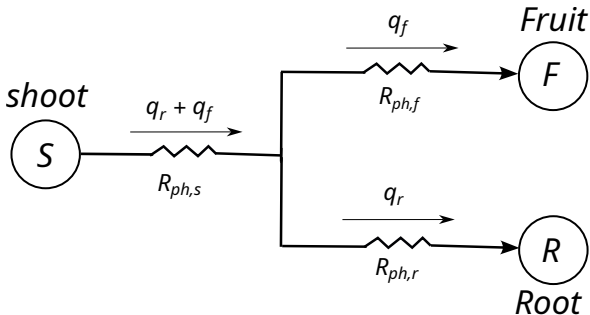
Carbon transport (T)



We get,

$$\begin{cases} (q_r + q_f)R_{ph,s} + q_r R_{ph,r} = (C_s - C_r) \\ (q_r + q_f)R_{ph,s} + q_f R_{ph,f} = (C_s - C_f) \end{cases}$$

Carbon transport (T)



We get,

$$\begin{cases} (q_r + q_f)R_{ph,s} + q_r R_{ph,r} = (C_s - C_r) \\ (q_r + q_f)R_{ph,s} + q_f R_{ph,f} = (C_s - C_f) \end{cases}$$

$$\begin{cases} T_r = q_r C_s \\ T_f = \left(\frac{W_s^n}{I^n + W_s^n} \right) q_f C_s \end{cases}$$

Water transport

Transpiration process A guides water flow, $A = \sigma_W W_s f(\psi_s)$,

$$\psi_r = \psi_{sol} - R_{sr}A, \quad \psi_s = \psi_r - R_{xy}A, \quad \psi_f = \psi_s.$$

where ψ_{sol} the soil water potential and $R_{..}$ resistances.

$$R_{sr} = \frac{r_{sr}}{W_r^{\alpha_r}}, \quad R_{xy,z} = \frac{r_{xy,z}}{W_z^{\alpha_z}}, \quad z = \{s, r\}.$$

Water regulation function

$$f(\psi) = \frac{\psi^n}{K^n + \psi^n}$$

